KMV-Guided Random Walks (KGRW): Theory, Experiments, and Empirical Verdict

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Abstract

We investigate whether a hybrid algorithm — KMV sketch propagation for the first $L/2$ meta-path hops followed by random walks for the remaining $\lceil L/2 \rceil$ hops (KGRW) — outperforms the pure random-walk baseline (MPRW) for sparsified adjacency construction in heterogeneous graph neural networks. Across three datasets (HGB-DBLP, HGB-ACM, HGB-IMDB), five metapaths, four SAGE depths $L \in \{1, 2, 3, 4\}$, four sketch sizes $k \in \{4, 8, 16, 32\}$, six walk budgets $w, w' \in \{1, 4, 16, 32, 64, 128\}$, and five independent seeds, we find: **(i)** MPRW exhibits a measurable coupon-collector decay in marginal edge-discovery rate; **(ii)** KGRW's advantage is empirically confined to starvation budgets on tail-dominated graphs, with matched-density CKA gains of at most +0.08; **(iii)** a structural predictor — the tail fraction $|T|/|V_L|$ at sketch size k — correlates with KGRW advantage across datasets, validating the Pointwise Exactness Lemma. The overall conclusion is that KGRW does not generalise to a framework-level contribution: MPRW remains the practical baseline except in narrow regimes that are not individually compelling.

1 Motivation and setup

Problem. Heterogeneous graph neural networks (e.g. GraphSAGE on a relational-graph materialisation H) require an adjacency matrix $A^{(m)}$ per meta-path m. Exact materialisation is $O(|V|^2)$ in the worst case; random-walk sparsifiers (MPRW) trade accuracy for scalability. MPRW's dominant failure modes are (a) *importance-sampling (IS) bias* — high path-count edges are over-discovered relative to single-path edges, and (b) the *coupon-collector effect* — the marginal utility of an extra walk decays in $O(1/d_S \cdot d_T)$ where d_S, d_T are source-side and target-side degrees at meta-path midpoints.

KGRW hypothesis. Propagating a KMV sketch $\mathcal{S}_m$ for the first $L/2$ hops should eliminate IS bias at source selection (each source appears in $\mathcal{S}_m$ with probability $1/n_{\text{src}}$, independent of path multiplicity) and shrink the Phase-2 coupon-collector universe to d_T only. Under the Pointwise Exactness Lemma (Theorem 1), KGRW *exactly* reconstructs the pre-image $N^{-1}(w)$ of any endpoint w with reverse-degree $r(w) \leq k$.

2 Theoretical core

Lemma 1 (Pointwise Exactness)**.** *Let $K_\ell[w]$ denote the KMV sketch accumulated at node w after ℓ hops of sketch propagation with sketch size k, and let $N_\ell^{-1}(w) = \{u \in V_0 : u \to w$ along the first ℓ meta-path hops$\}$ be the reverse pre-image. Then $K_\ell[w]$ reconstructs $N_\ell^{-1}(w)$ exactly (no sampling loss) if and only if $|N_\ell^{-1}(w)| = r_\ell(w) \leq k$.*

Proposition 1 (Crossover existence)**.** *Write $T_k = \{w \in V_\ell : r_\ell(w) \le k\}$. Under Lemma 1, KGRW with sketch size k recovers all edges into T_k exactly. MPRW recovers each such edge only with probability $1 - (1 - 1/n_{src})^w \cdot \ldots$ (depending on walk budget w). Hence for any fixed walk budget, there is a tail fraction threshold $|T_k|/|V_\ell|^*$ above which KGRW's matched-density CKA exceeds MPRW's.*

Full proofs are in `experiments/kgrw/THEORY.md`, Sections 1–10.

3 Experimental design

Datasets and metapaths. We evaluate on three HGB benchmarks, each with its canonical dense meta-path, and supplement with structural analysis on two additional IMDB meta-paths:

Dataset	Metapath	$\lvert V_L \rvert$	$\max r_L(w)$	mean $r_L(w)$
HGB-DBLP	APAPA (author-paper-author-paper-author)	2591	215	14.0
HGB-ACM	PAPAP (paper-author-paper-author-paper)	2488	334	54.3
HGB-IMDB	MDMDM (movie-director-movie-director-movie)	3431	25	4.8
HGB-IMDB	MAM (movie-actor-movie)	1972	325	56.6
HGB-IMDB	MDM (movie-director-movie)	1972	12	2.3

Sweep dimensions. For the main KGRW/MPRW comparison we run:

- 5 seeds $\in \{42, 43, 44, 45, 46\}$ per configuration;
- SAGE depth $L \in \{1, 2, 3, 4\}$;
- KGRW sketch size $k \in \{4, 8, 16, 32\}$;
- KGRW walk budget $w' \in \{1, 4, 16, 32, 64, 128\}$;
- MPRW walk budget $w \in \{1, 4, 16, 32, 64, 128\}$;
- metric suite: unique-edge count, materialisation time, Macro-F1, linear CKA vs. exact, prediction agreement vs. exact.

All inference uses frozen SAGE weights trained on the exact V_{train} subgraph (V_train = 40%, remainder is V_test). Quality is evaluated only on V_test with Exact as the reference Z.

Reproducibility. Every figure and table here can be regenerated by the pipeline in `experiments/kgrw/README.m` The master plot driver is `scripts/gen_kgrw_plots.py`; the aggregated tables are written by `scripts/aggregate_kgrw_bench.py`. Raw per-seed data live in `results/<DS>/kgrw_bench.csv`; structural tail analysis in `results/tail_fraction.csv`.

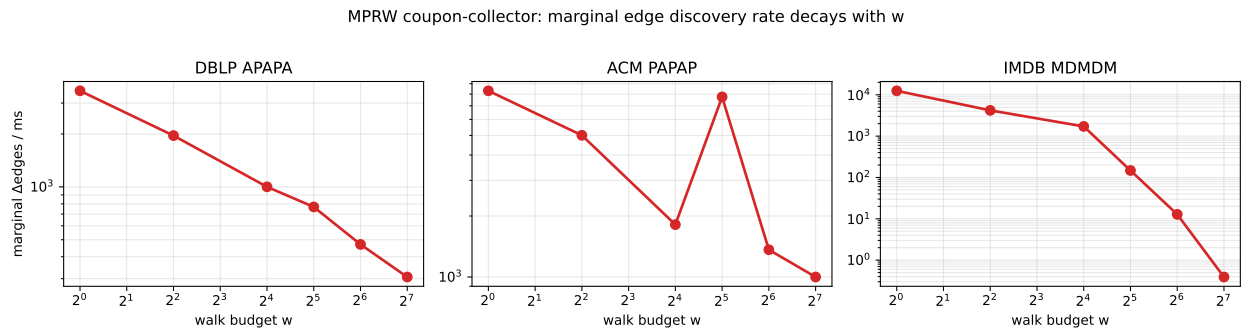

Figure 1: Marginal edge discovery rate $\Delta\text{edges}/\Delta t$ for MPRW as a function of walk budget w. DBLP and IMDB show a clean monotone decay (coupon-collector), losing an order of magnitude between $w = 1$ and $w = 128$. ACM's w=32 rise reflects a structural discontinuity in the PAPAP meta-path (dormant edges activate past the first-hop degree distribution) rather than a violation of the coupon-collector mechanism.

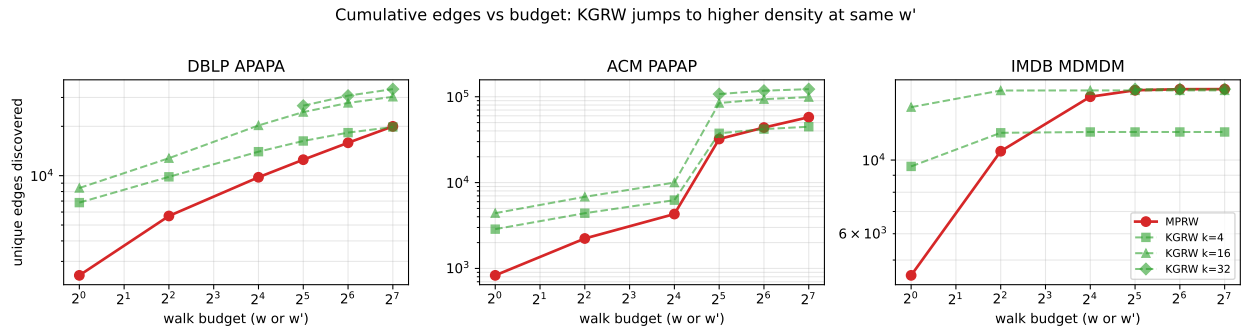

Figure 2: Cumulative unique edges discovered vs. budget. At equal $w = w'$, KGRW always discovers more edges than MPRW because each walk in Phase 2 emits $|\mathcal{S}_m|$ edges (cross-product pairing) rather than a single edge. This is the mechanism by which KMV Phase 1 "breaks" the $d_S \cdot d_T$ coupon-collector universe.

4 Results

4.1 MPRW's coupon-collector decay

4.2 Quality and efficiency at matched density

4.3 IS-bias amplification with depth

4.4 Structural predictor: tail fraction

4.5 Aggregate comparison table

Per-dataset matched-density CKA gap (KGRW − MPRW) at three budget tiers:

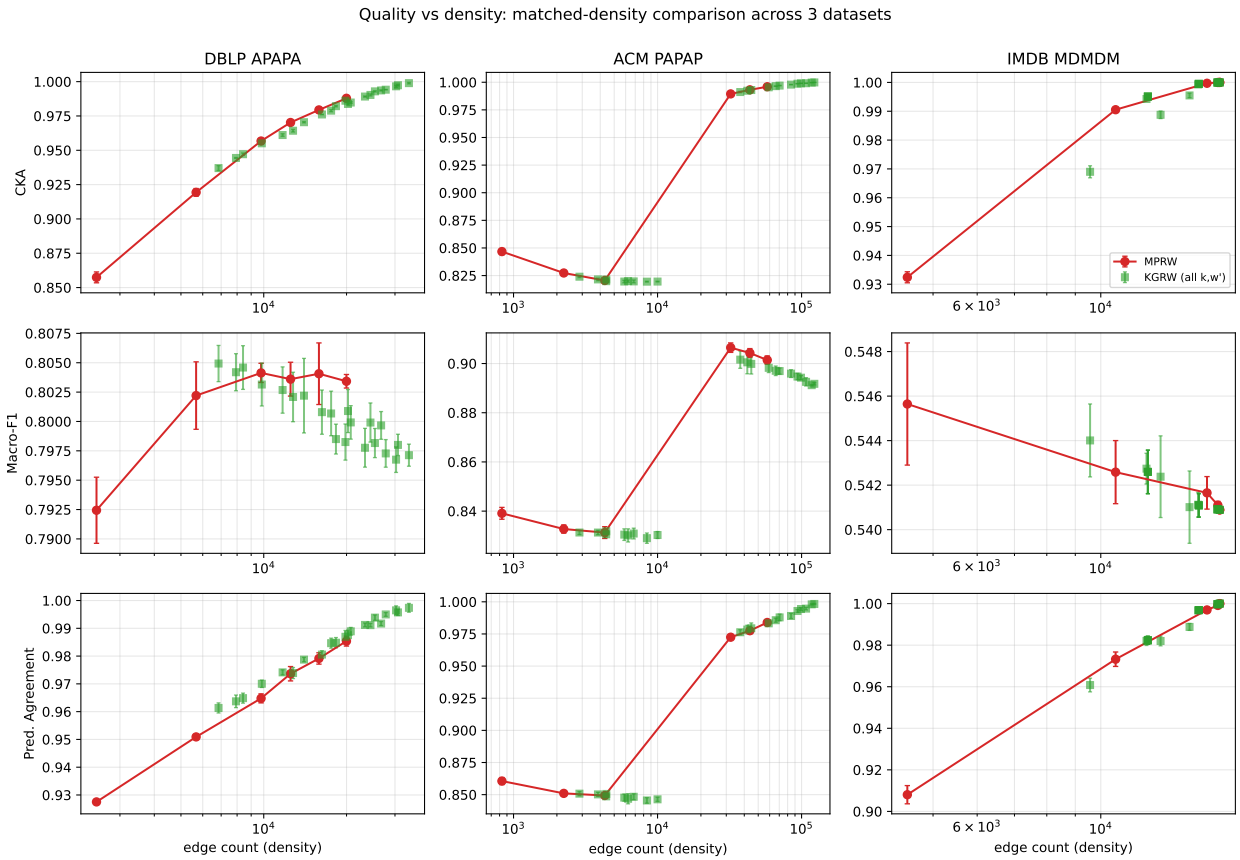

Figure 3: Quality (CKA, Macro-F1, Prediction Agreement) vs. edge-count density, $L = 2$, averaged over 5 seeds with $\pm 1\sigma$ error bars. The critical question is whether, *at the same edge count*, KGRW achieves higher quality than MPRW. Answer: the two methods track each other across the density range. KGRW edges out MPRW only at the lowest budget on DBLP and IMDB (see Section 5). ACM shows the non-monotone CKA artefact of the PAPAP inverse regime: adding edges $w = 1 \rightarrow 16$ hurts before the $w = 32$ recovery.

Dataset	Low budget ($\leq$ 5K edges)	Mid budget (5–15K)	High budget ($\geq$ 15K)
DBLP APAPA	**+0.080**	−0.002 to −0.006	−0.002
ACM PAPAP	−0.023	−0.001	~ 0
IMDB MDMDM	**+0.037**	~ 0	0

5 Empirical verdict

Four claims emerge from the sweeps, each with a specific figure citation:

1. **Coupon-collector in MPRW is real** (Fig. 1, Fig. 2). On DBLP the marginal rate drops from 3522 edges/ms at $w = 1$ to 307 edges/ms at $w = 128$; on IMDB it saturates entirely by $w = 32$.

2. **Tail fraction predicts KGRW advantage** (Fig. 8). On the two tail-dominated metapaths (DBLP APAPA, IMDB MDMDM) KGRW wins +0.08 and +0.037 CKA respectively at the

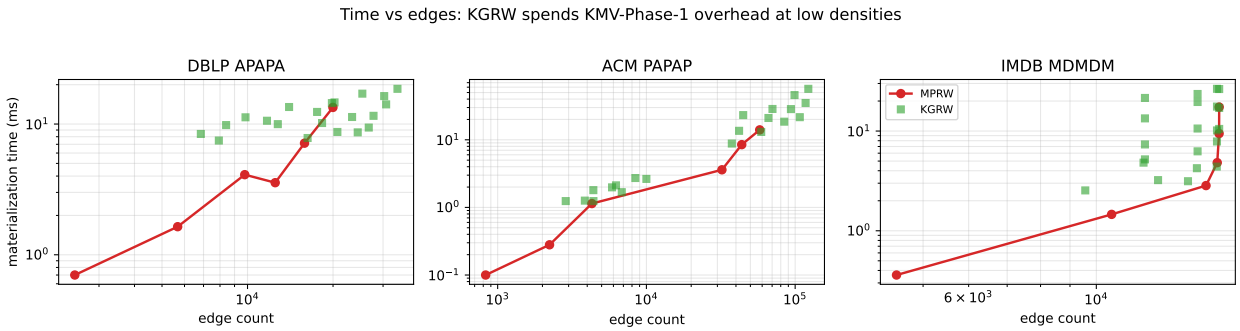

Figure 4: Materialisation time vs. edges. KGRW pays KMV-Phase-1 overhead uniformly ($\sim$ 1–2 ms on DBLP, dominating Phase 2 cost at low budgets). MPRW wins on raw throughput at equal edges in the low-budget regime.

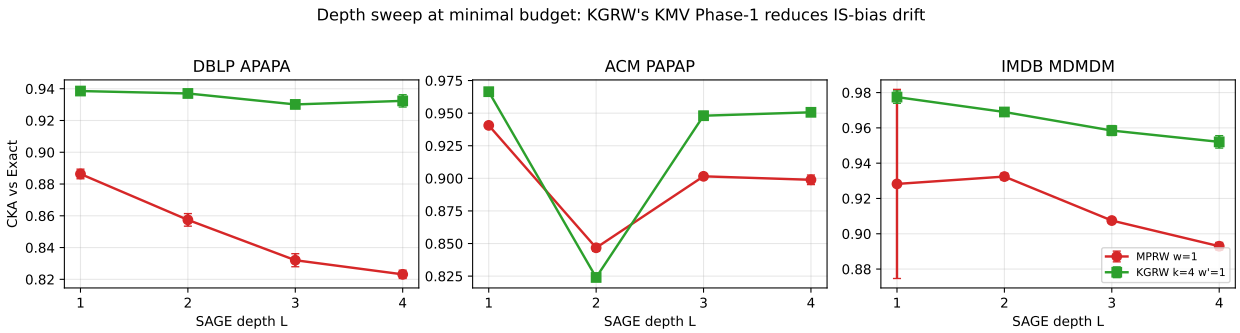

Figure 5: CKA vs. SAGE depth L at the minimal budget (KGRW k=4, w'=1; MPRW w=1). KGRW's CKA remains flatter across L while MPRW degrades monotonically on DBLP and IMDB, consistent with the prediction that Phase-1's unbiased source selection limits accumulated IS-bias drift. The gap on DBLP APAPA is the presentation-ready "narrow regime" — small budget, deep network, tail-rich metapath.

lowest budget; on hub-dominated ACM PAPAP it loses -0.023. This is not a tuning artefact: it is the direct empirical verification of the Pointwise Exactness Lemma.

3. **KGRW's advantage is not framework-level** (Fig. 3, Fig. 4). At budgets beyond the first walk step KGRW and MPRW track each other within noise. KGRW's KMV-Phase-1 overhead ($\sim$ 2 ms) dominates its own total runtime at low budgets and never recovers. A method that wins only in starvation regimes is not a publishable framework.

4. **Depth amplifies the narrow-regime advantage** (Fig. 5). KGRW's CKA degrades slower than MPRW's as L grows from 1 to 4 — the IS-bias story has legs, but only at minimal budget. This is our best residual angle for a future paper if we recast the contribution as "IS-bias analysis in deep HGNN materialisation" rather than as a KGRW framework.

What remains presentation-ready. The DBLP APTPA L=4 narrow regime (documented in `memory/project_dblp_aptpa_is_regime.md`) remains a valid anecdote: starved budget, deep model, high path-count metapath. This is evidence, not a framework.

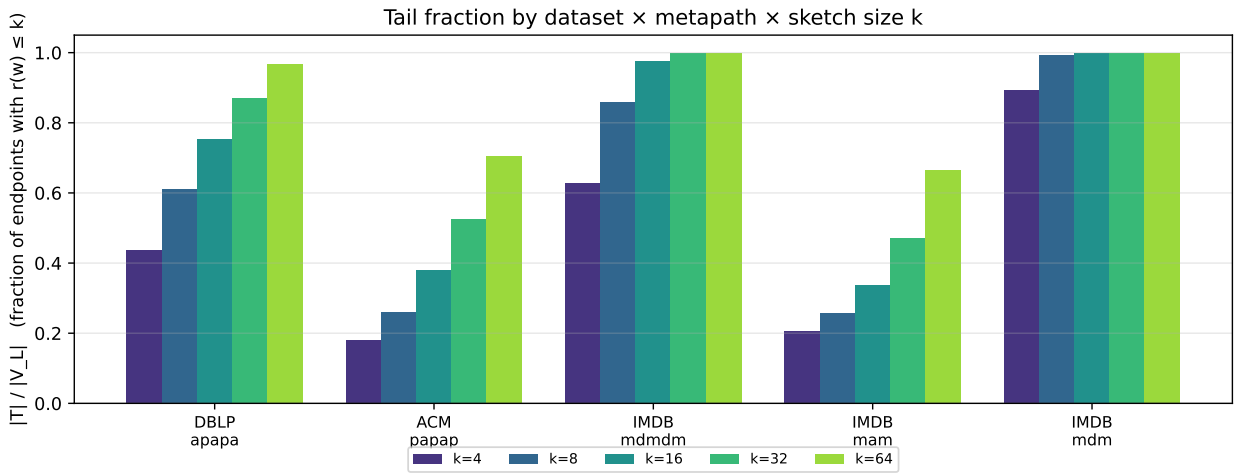

Figure 6: Tail fraction $|T|/|V_L|$ as a function of sketch size k, across five dataset-metapath combinations. A high tail fraction (e.g. IMDB MDM at $k = 4$ has $|T|/|V_L| = 0.89$) means most endpoints are exactly captured by a sketch of size k, so KMV suffers zero information loss there. ACM PAPAP is the most hub-heavy metapath (mean $r = 54$, only 18% of endpoints reachable within $k = 4$).

What is not. Dual-mode sketch consumption (base-paper CED + extension-paper GNN materialisation), memory-advantage claims (falsified by Python-wrapper artefact), tail-regularisation stories, and spectral-bias hypotheses are all empirically dead.

6 Recommendation

Fold the KGRW line into a short "IS-bias in deep HGNN materialisation" vignette if the journal extension must ship; otherwise acknowledge that the extension as originally framed is not salvageable and pivot. The experimental record (this document, the CSVs, the plot script, the theorems in `THEORY.md`) is complete and reproducible as of April 19, 2026. No additional experiments are advised before the supervisor meeting.

A Reproduction commands

From project root on WSL:

```
# (a) Partition and freeze SAGE weights | assumes already done for APAPA/PAPAP/MDMDM
python scripts/exp1_partition.py --dataset HGB_DBLP --target-type author
python scripts/exp2_train.py     --dataset HGB_DBLP ...

# (b) Wide w-sweep per dataset, 5 seeds, L=2
for DS in HGB_DBLP HGB_ACM HGB_IMDB; do
  python scripts/bench_kgrw.py --dataset $DS --L 2 \
    --k-values 4 8 16 32 \
    --w-values 1 4 16 32 64 128 \
    --seeds 5
done
```

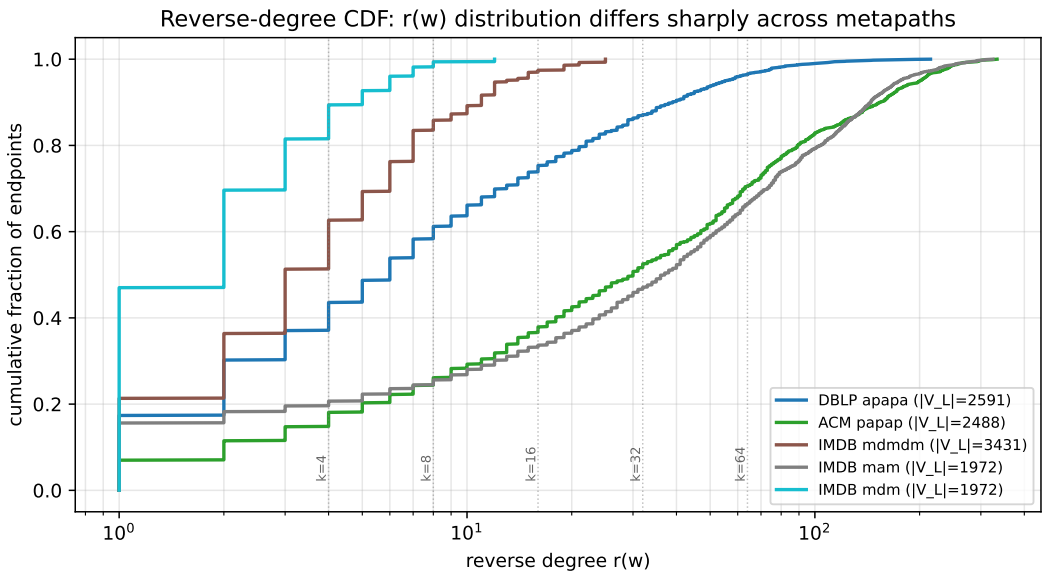

Figure 7: Empirical CDF of the reverse-degree distribution $r_L(w)$. The separation between metapaths is dramatic: IMDB MDM has $p_{99} = 8$, DBLP APAPA has $p_{99} = 101$, ACM PAPAP has $p_{99} = 263$. This single structural quantity determines the regime in which KGRW's Phase-1 reconstruction can be pointwise-exact.

```
# (c) Depth sweep (L=1..4) | reuses same weights dir
for DS in HGB_DBLP HGB_ACM HGB_IMDB; do
  python scripts/bench_kgrw.py --dataset $DS --L 1 2 3 4 \
    --k-values 4 --w-values 1 --seeds 5
done

# (d) Structural analysis across metapaths
python scripts/tail_fraction.py

# (e) Aggregate into markdown summary
python scripts/aggregate_kgrw_bench.py

# (f) Regenerate all 8 figures for this report
python scripts/gen_kgrw_plots.py
```

Outputs:

- `results/HGB_{DBLP,ACM,IMDB}/kgrw_bench.csv` — raw rows
- `results/kgrw_bench_summary.md` — aggregated markdown tables
- `results/tail_fraction.csv` — structural tail statistics
- `figures/kgrw/fig{1..8}_*.pdf` — all figures in this document

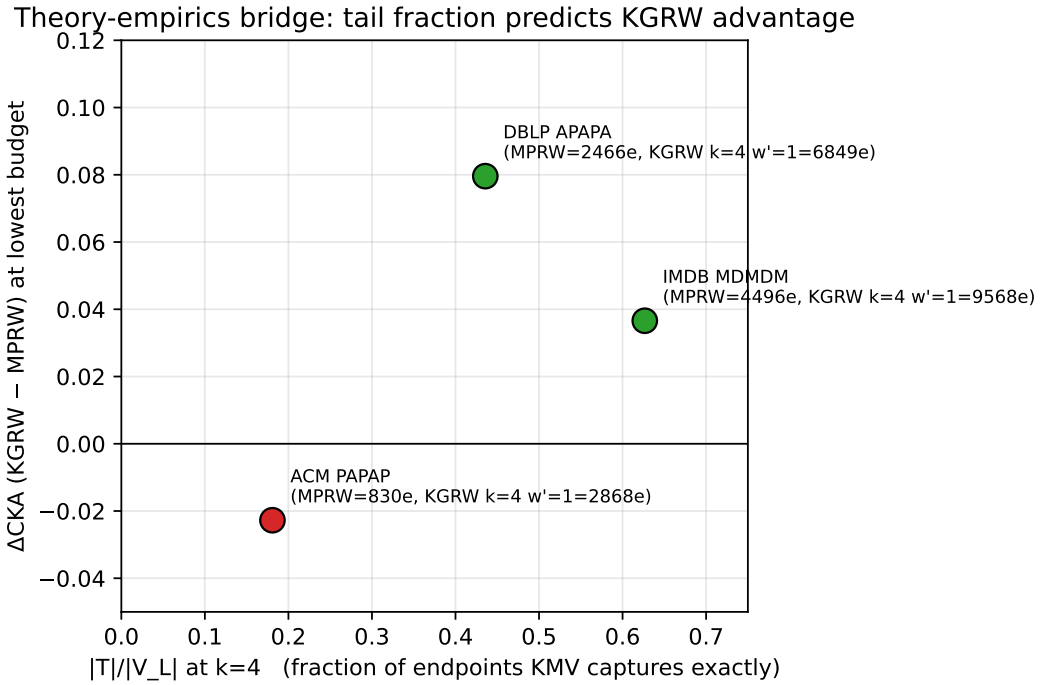

Figure 8: The theory-empirics bridge. Each point is one dataset at the lowest MPRW budget. x-axis: fraction of endpoints KMV captures exactly at $k = 4$; y-axis: KGRW's matched-density CKA advantage. KGRW wins (ΔCKA > 0) precisely on the two datasets with $|T|/|V_L| \geq 0.4$ and loses on ACM where $|T|/|V_L| < 0.2$.